



Smart Student Information System Assistant

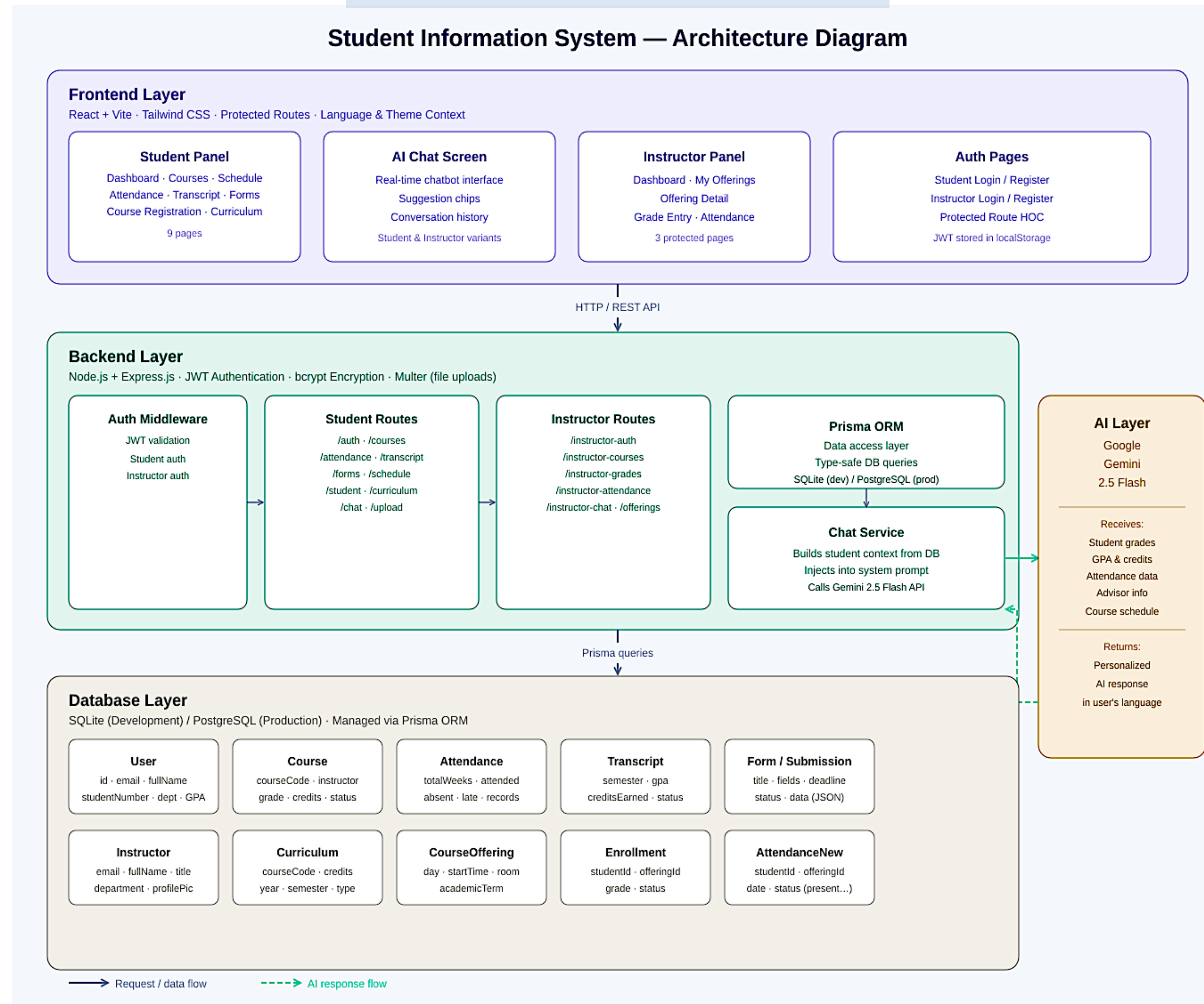
Mithat ALTAY, Zeynep AYDEMİR, Ayberk Fevzi REÇBER

Dr. Assoc. Prof. Şadi ŞEHAB

Software Engineering



SYSTEM ARCHITECTURE

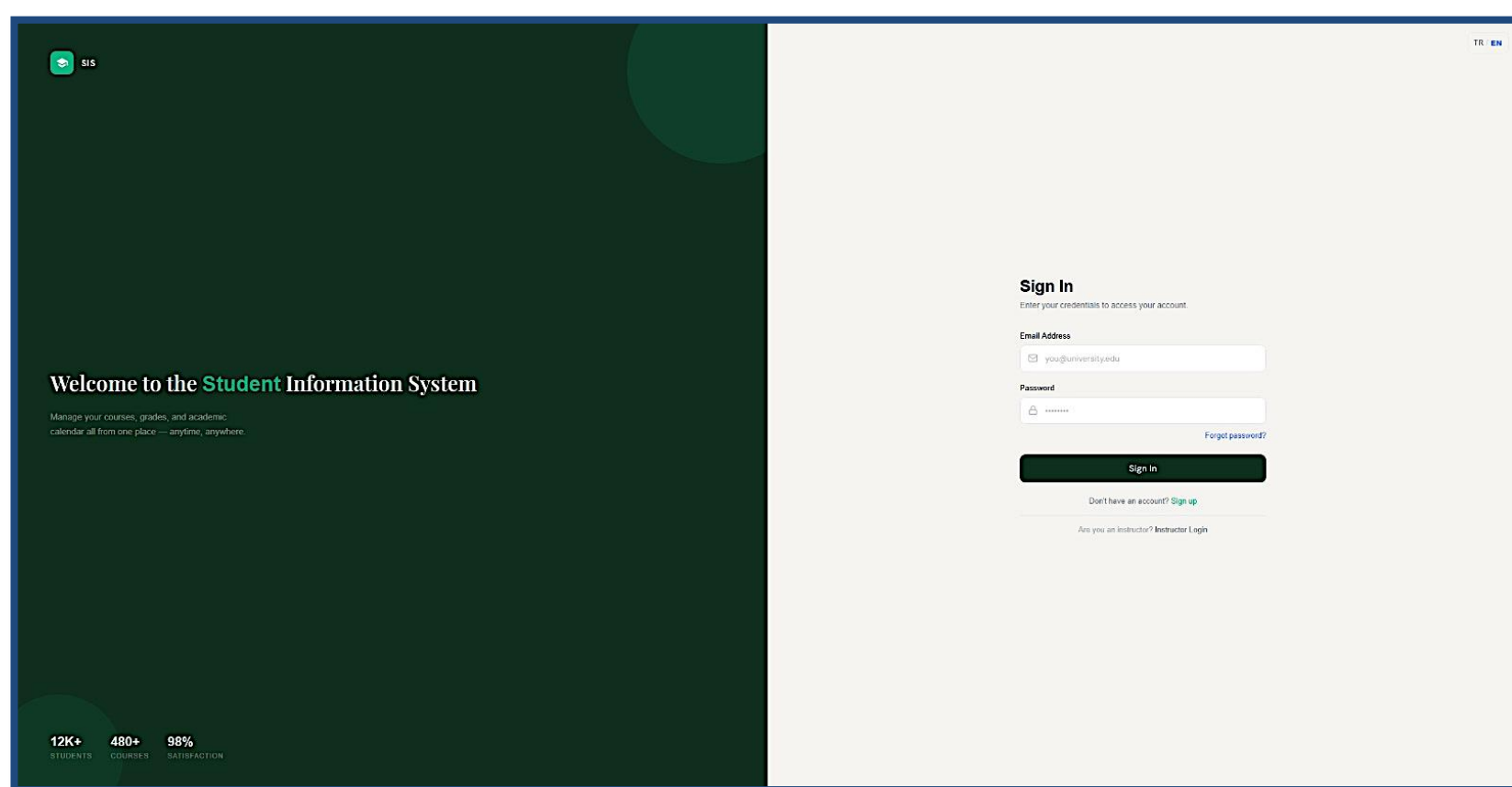


APPLICATION FEATURES

Student Modules:

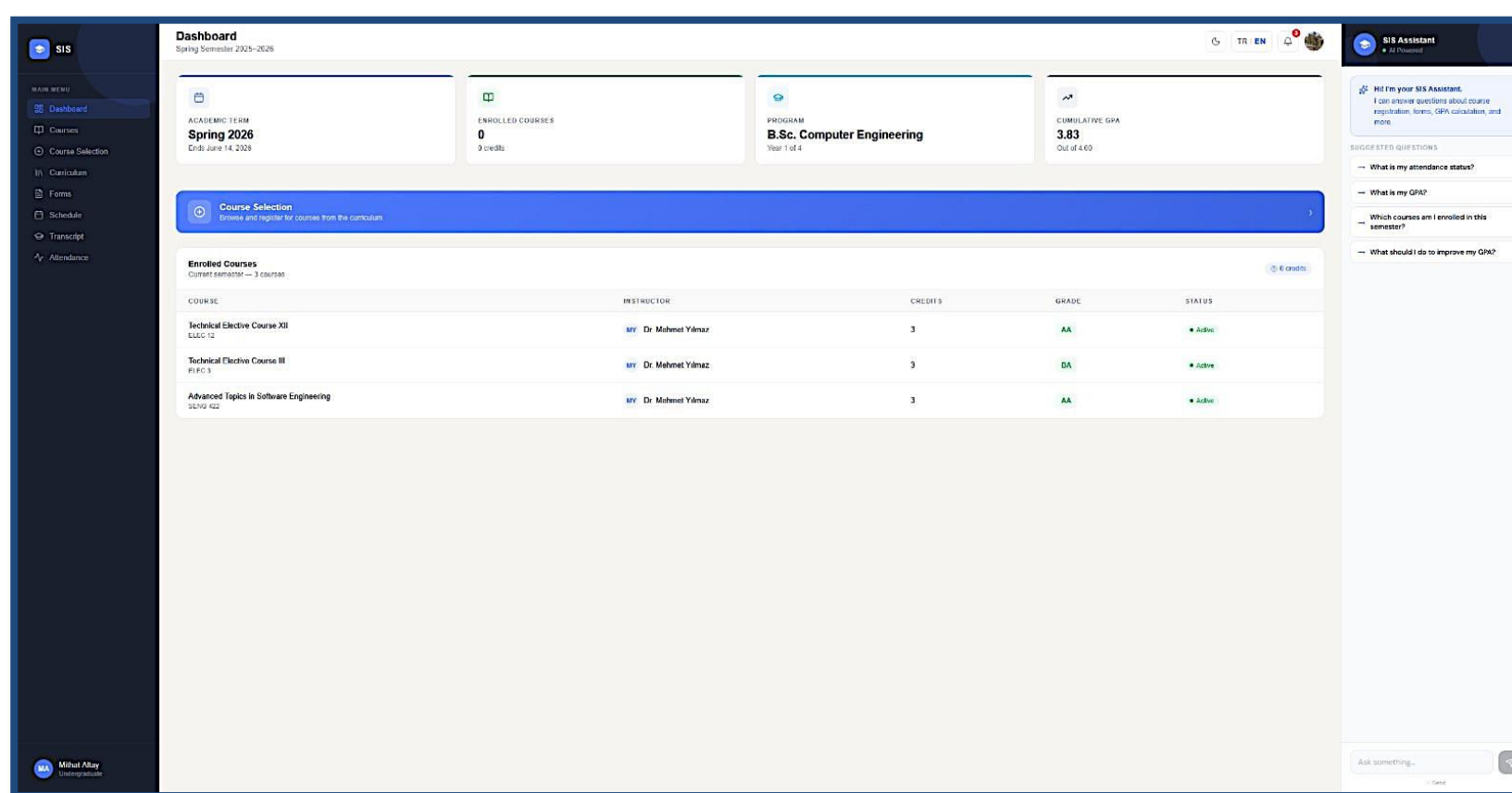
- Registration & Login: Account creation with email/password, secure authentication via JWT
 - Dashboard: GPA, total credits, and semester overview in one screen
 - Courses: Active courses with grade and progress tracking
 - Course Schedule: Weekly calendar with automatic daily filtering
 - Attendance: Weekly records, percentage calculation, and warnings
 - Transcript: Semester-based grades, cumulative GPA, credit summary
 - Course Registration: View available courses and manage enrollment
 - AI Assistant: Personalized chatbot using real academic data
- ### Instructor Modules:
- Course Management: Add/remove and manage courses
 - Student Tracking: View enrolled students and their details
 - Grade Entry: Enter grades with automatic GPA calculation
 - Attendance: Record weekly attendance and track absences

Sign In Page



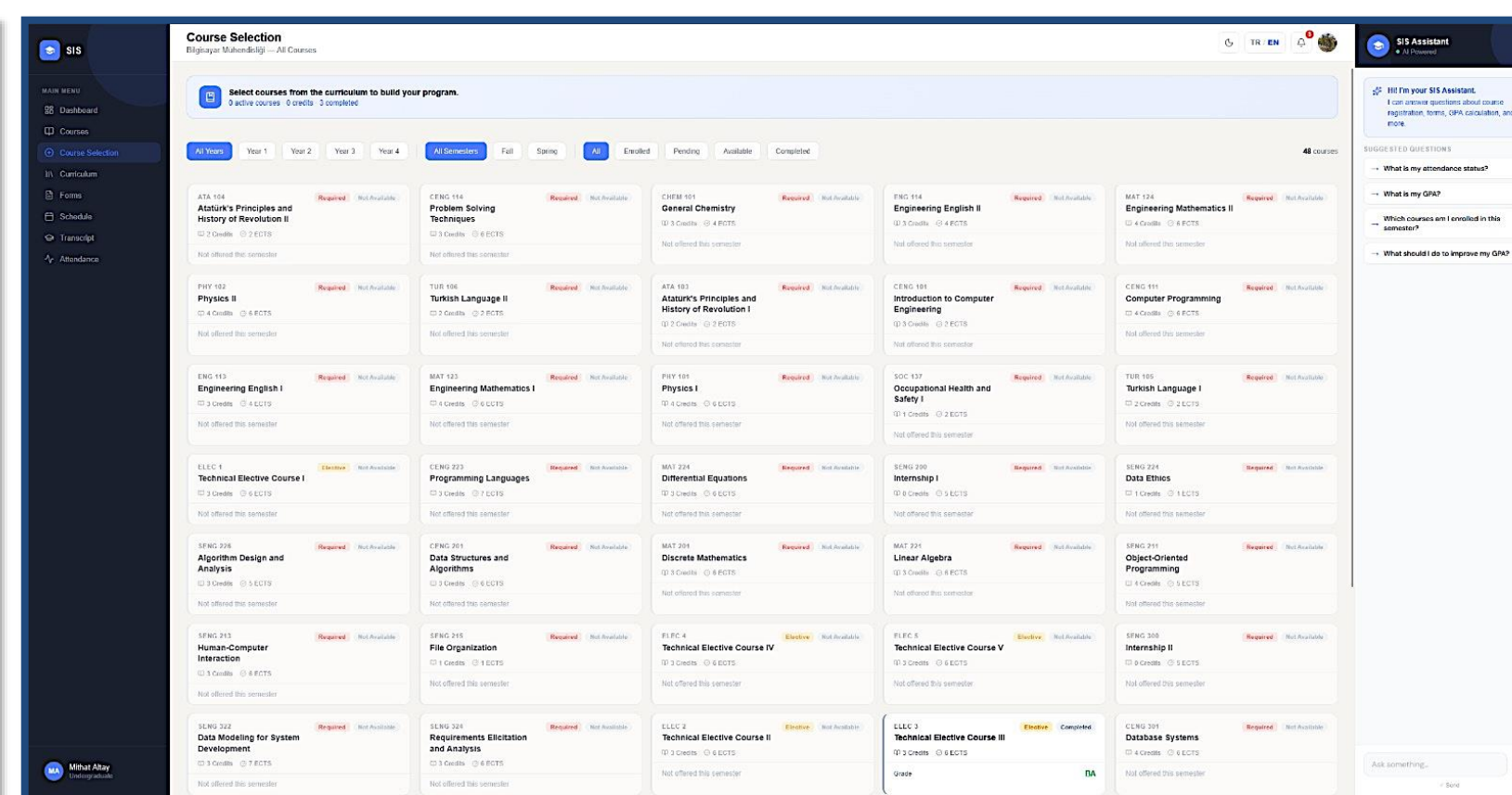
Secure JWT-based authentication for students and instructors.

Student Dashboard



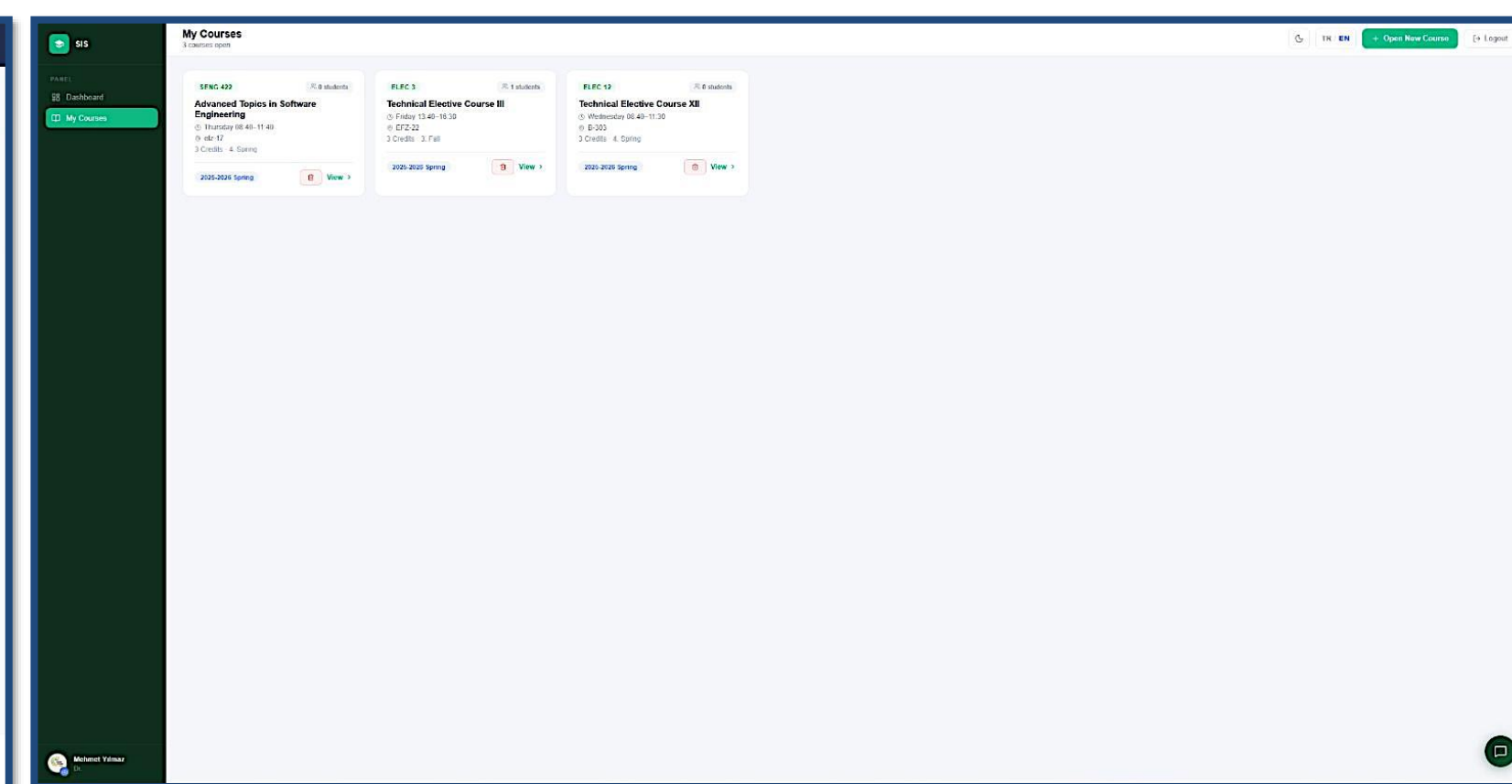
Personalized overview of courses, GPA, attendance and upcoming classes.

Courses Page

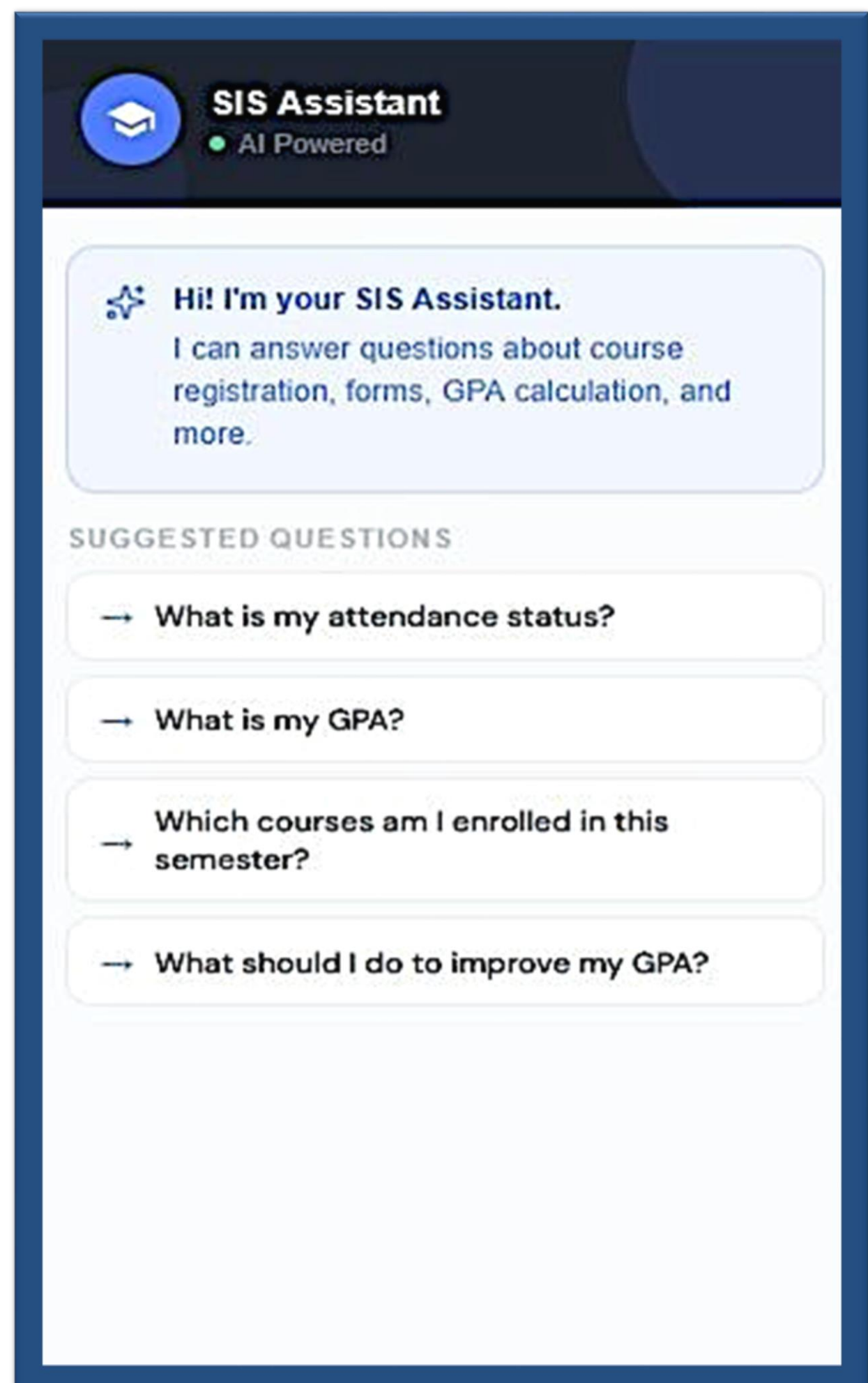


Full course list with schedule, instructor, room and live progress tracking.

Instructor Courses Page

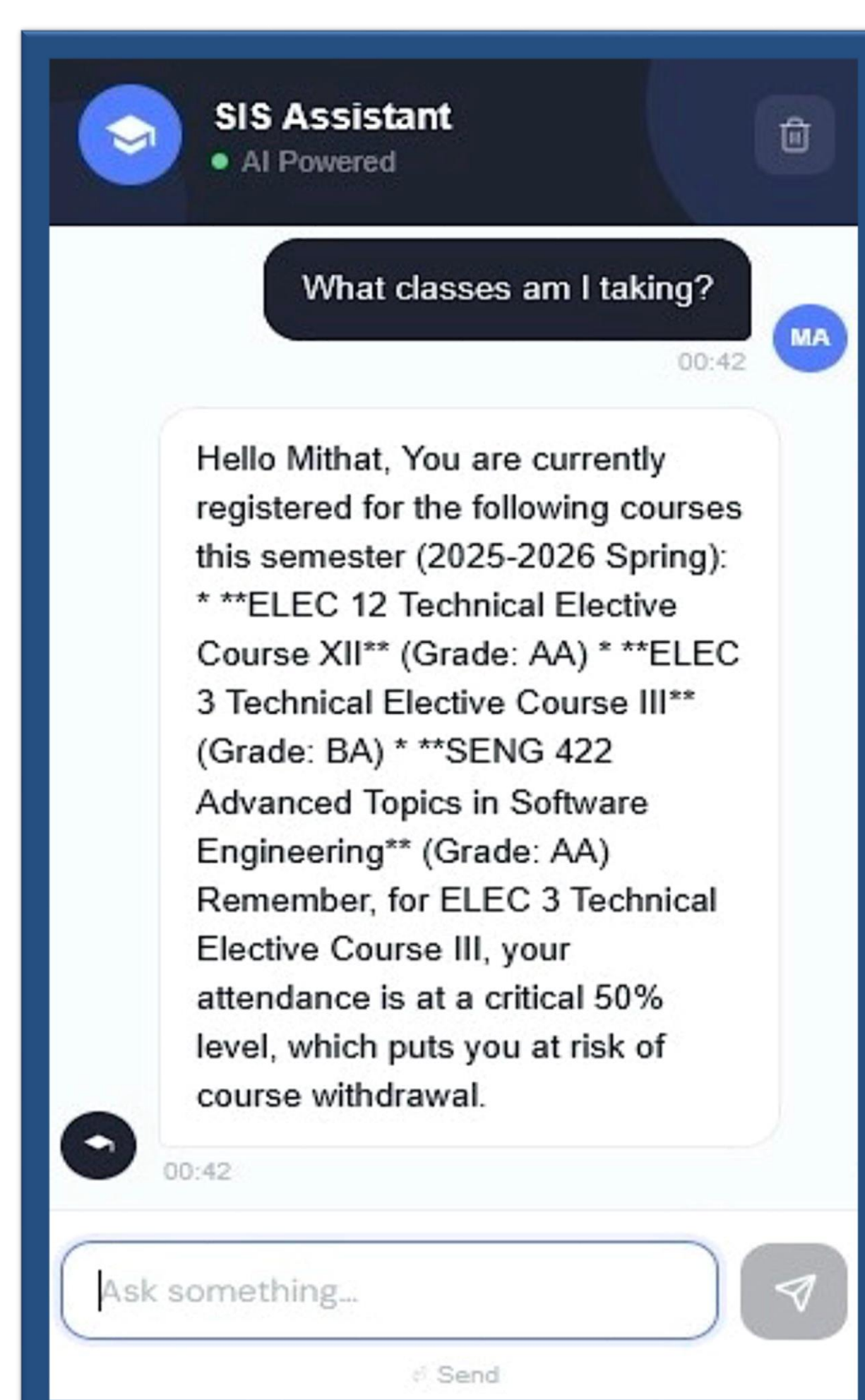


Manage offerings, view enrolled students, enter grades and record attendance.



How It Works

- User sends a natural language message
- Backend authenticates the request via JWT
- Student / instructor data is fetched from the database
- Data is injected into the AI system prompt
- Gemini 2.5 Flash generates a personalized response
- Response is returned to the user in real time



Sample Interactions

- "What is my current GPA?" → Instantly retrieved from transcript data
- "Do I have any classes tomorrow?" → Pulled from live schedule
- "Which courses am I at risk of failing due to absences?" → Cross-referenced with attendance records
- "What is the deadline for grade submissions?" → Answered from instructor context

LAYER	TECHNOLOGY	DESCRIPTION	RESULTS AND CONTRIBUTIONS	TOOL	PURPOSE	REFERENCES
Frontend	React + Vite	User interface development	A fully functional Student Information System suitable for real university use has been successfully developed within the scope of this project. - Separate, role-based panels for students and instructors were created, simulating the structure of a real university information system. - All core SIS modules (courses, grades, attendance, transcript, course registration, petitions) have been fully implemented. - The AI assistant — the project's most original contribution — works with the student's real academic data to produce personalized, data-driven responses, transforming a conventional SIS into an intelligent academic advisor. - An industry-standard secure authentication infrastructure was established using JWT and bcrypt. - A modern, user-friendly, and responsive interface was built.	Visual Studio Code	Code development environment	Google. (2024). <i>Gemini 2.5 Flash API Documentation</i> . Google AI Studio.
Styling	Tailwind CSS	Responsive design		Postman	API testing and documentation	
Backend	Node.js + Express.js	Server and API management		Git / GitHub	Version control and code management	Prisma. (2024). <i>Prisma ORM Documentation</i> . https://www.prisma.io/docs
Database ORM	Prisma ORM	Database querying and management		Node Package Manager (npm)	Package management	Meta Open Source. (2024). <i>React Documentation</i> . https://react.dev
Database	SQLite / PostgreSQL	Data storage		Prisma Studio	Visual database management	OpenJS Foundation. (2024). <i>Node.js Documentation</i> . https://nodejs.org/docs
Authentication	JWT + bcrypt	Secure session and encryption		Google AI Studio	Gemini API key management and testing	Auth0. (2023). <i>JSON Web Tokens Introduction</i> . https://jwt.io/introduction
Artificial Intelligence	Google Gemini 2.5 Flash	Natural language processing and response generation		Figma	UI design prototyping	Tailwind Labs. (2024). <i>Tailwind CSS Documentation</i> . https://tailwindcss.com/docs